ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

ENGINEERING DRAWING AND CAD

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 1st year Engineering Drawing and CAD paper 1 · practice a.

Written by Academic Central for practice, following first-year Engineering Drawing and CAD (graphics) syllabuses at South African universities, following SANS 10111 engineering drawing practice. It is not a university memorandum.

How to mark: each ✓ is one mark. Accept any correct answer in the student's own words. For QUESTION 2, give marks for each correct feature of a view even if another feature is wrong. Deduct marks for views placed in third-angle positions only once.

QUESTION 1

1.1 Any FOUR: (4)
Continuous thick: visible outlines and edges ✓
Continuous thin: dimension lines, projection lines, hatching ✓
Dashed thin: hidden outlines and edges ✓
Chain thin: centre lines and lines of symmetry ✓
Chain thin, thick at the ends and at changes of direction: cutting planes

1.2 TL = $\sqrt{(40^2 + 30^2)}$ ✓ = 50 mm ✓ (3)
By drawing: rotate the line (or use an auxiliary view) until it is parallel to the plane of projection; it then shows in its true length ✓

1.3.1 240 ÷ 5 = 48 mm ✓ (1)

1.3.2 15 × 2 = 30 mm ✓ (1)

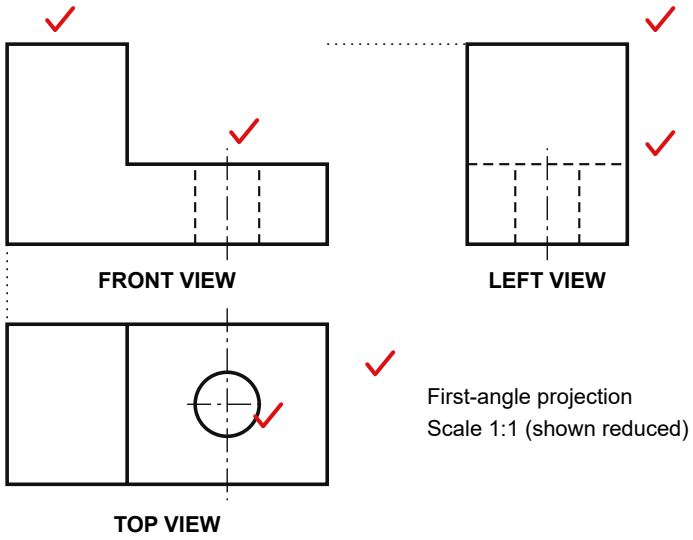
1.4 Any FOUR: title of the drawing; drawing number; scale; projection method (symbol); name of the person who drew it; date; units; material; company or institution name; sheet number ✓✓✓✓ (4)

1.5 First angle: the top view is placed BELOW the front view and the left view is placed on the RIGHT of the front view ✓ (2)
Third angle: the top view is placed ABOVE the front view and the left view is placed on the LEFT ✓

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QUESTION 2

2.1



Front view: L-shaped outline 80 × 50 with the step at 30 and 20 ✓✓✓✓; hidden lines of the hole ✓✓; correct size ✓; placed top left ✓. Top view: 80 × 40 rectangle ✓✓; line at the step (30 from the left) ✓✓; Ø16 circle in the right place ✓✓; placed below the front view ✓✓. Left view: 40 × 50 rectangle ✓✓; hidden line at height 20 ✓; hidden lines of the hole ✓; placed to the right of the front view ✓✓

(22)

2.2 Hidden lines dashed and thin in the front and left views ✓✓ (4)
Centre lines (chain lines) through the hole in all three views ✓✓

2.3 First-angle symbol drawn correctly (truncated cone: side view on the left of the end view) ✓ (4)
Any THREE correct main dimensions, e.g. 80 length, 50 height, 40 depth, Ø16 hole, 55 to the hole centre ✓✓✓

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QUESTION 3

3.1 A sectional view shows the inside shape of a part clearly, without many confusing hidden lines ✓✓ (4)
Hatching rules (any TWO): thin lines at 45°; evenly spaced (about 3–4 mm); different parts hatched in different directions or spacing; the same part hatched the same way in all views ✓✓

3.2 Any THREE: shafts; bolts, screws and studs; nuts and washers; keys and pins; ribs and webs; spokes; rivets; ball bearings ✓✓✓ (3)

3.3 Any FIVE: each dimension is shown once only; dimensions are placed outside the view where possible; dimension lines do not cross other lines where it can be avoided; hidden lines are not dimensioned; centre lines are not used as dimension lines; units are stated once in the title block; arrowheads touch the projection lines; projection lines leave a small gap from the outline; the smallest dimensions are placed closest to the view ✓✓✓✓✓ (5)

3.4 M: ISO metric thread ✓ 12: nominal (major) diameter of 12 mm ✓ 1.75: pitch of 1.75 mm ✓ (3)

3.5 Chain: each dimension is measured from the end of the previous one ✓
Parallel: all dimensions are measured from one datum ✓
Parallel is more accurate because errors (tolerances) do not add up along the chain ✓ (3)

3.6 Aligned: dimensions are written parallel to the dimension line and read from the bottom or the right of the sheet ✓
Unidirectional: all dimensions are written horizontally and read from the bottom of the sheet ✓ (2)

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QUESTION 4

4.1 Hole: upper limit 25.021 ✓, lower limit 25.000 ✓
Shaft: upper limit 24.993 ✓, lower limit 24.980 ✓ (4)

4.2 Maximum clearance = largest hole − smallest shaft = 25.021 − 24.980 ✓ = 0.041 mm ✓
Minimum clearance = smallest hole − largest shaft = 25.000 − 24.993 ✓ = 0.007 mm ✓ (4)

4.3 Clearance fit (a close running or sliding fit) ✓ (1)

4.4 (a) casting ✓ (b) forging ✓ (c) extrusion ✓ (3)

4.5 (a) drilling ✓ (b) turning ✓ (c) milling ✓ (3)

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QUESTION 5

5.1 Any FOUR: faster to draw and change; more accurate; drawings are easy to copy, store and send electronically; parts and symbols can be reused from libraries; 3D models can be viewed from any angle; the model can be used for analysis, simulation, CNC machining or 3D printing; dimensions and parts lists can be produced automatically ✓✓✓✓ (4)

5.2 2D drafting: lines, arcs and text on a flat drawing, like drawing by hand on a computer ✓
3D solid modelling: a full solid with volume and mass, from which views can be produced ✓
Parametric modelling: the model is driven by dimensions and constraints ✓; changing a dimension updates the model and its drawings automatically ✓ (4)

5.3 Sketch the L-shaped side profile (the front face) on the front plane with dimensions 80, 50, 30 and 20 ✓✓ (6)
Extrude the sketch 40 mm to give the depth ✓
Sketch a Ø16 circle on the top face of the base, 55 from the left end and 20 from the front ✓
Cut-extrude the circle "through all" to make the hole ✓
Check the model and produce the drawing views from it ✓
(Accept other correct orders, e.g. extrude the base and add the upright as a second extrusion.)

5.4.1 (20 + 30, 30 + 40) ✓ = (50, 70) ✓ (2)

5.4.2 50 units at 90° is straight up ✓, so the end point is (20, 80) ✓ (2)

5.5 STEP: a neutral format for exchanging exact 3D CAD models between different CAD programs ✓ (2)
STL: a triangle (mesh) model, used mainly for 3D printing ✓

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TOTAL: 100

END OF MARKING GUIDELINE